English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

# Probability and Statistics, Computer Science

Test 2, 12 June 2022: solutions

*English transcription of the supplied handwritten solution notes.*

## Problem 1

### 1. Normalisation and moments

$$\begin{aligned} 1 &= \int_0^1 \int_0^{1-x} (cx+1)\,dy\,dx = \int_0^1 (cx+1)(1-x)\,dx \\ &= \frac{c}{2} + 1 - \frac{c}{3} - \frac{1}{2} = \frac{c}{6} + \frac{1}{2}, \end{aligned}$$

so $c = 3$. Also

$$\begin{aligned} \mathbb{E}[XY] &= \int_0^1 \int_0^{1-x} xy(cx+1)\,dy\,dx = \int_0^1 x(cx+1)\frac{(1-x)^2}{2}\,dx \\ &= \frac{c}{2}\left(\frac{1}{3} - \frac{2}{4} + \frac{1}{5}\right) + \frac{1}{2}\left(\frac{1}{2} - \frac{2}{3} + \frac{1}{4}\right) = \frac{c}{60} + \frac{1}{24} = \frac{11}{120}. \end{aligned}$$

For $0 \le x \le 1$, $f_X(x) = (cx+1)(1-x)$, hence

$$\mathbb{E}X = \int_0^1 (cx^2 + x)(1-x)\,dx = \frac{c}{3} + \frac{1}{2} - \frac{c}{4} - \frac{1}{3} = \frac{c}{12} + \frac{1}{6} = \frac{5}{12}.$$

Similarly,

$$\mathbb{E}X^2 = \int_0^1 (cx^3 + x^2 - cx^4 - x^3)\,dx = c\left(\frac{1}{4} - \frac{1}{5}\right) + \frac{1}{3} - \frac{1}{4} = \frac{c}{20} + \frac{1}{12} = \frac{7}{30},$$

$$DX = \frac{7}{30} - \left(\frac{5}{12}\right)^2 = \frac{43}{720} \approx 0.0597.$$

For $0 \le y \le 1$,

$$f_Y(y) = \int_0^{1-y} (cx+1)\,dx = \frac{c}{2}(1-y)^2 + 1 - y.$$

Therefore

$$\begin{aligned} \mathbb{E}Y &= \frac{c}{2}\int_0^1 y(1-y)^2\,dy + \int_0^1 y(1-y)\,dy \\ &= \frac{c}{2}\left(\frac{1}{2} - \frac{2}{3} + \frac{1}{4}\right) + \frac{1}{2} - \frac{1}{3} = \frac{c}{24} + \frac{1}{6} = \frac{7}{24}, \\ \mathbb{E}Y^2 &= \frac{c}{2}\left(\frac{1}{3} - \frac{2}{4} + \frac{1}{5}\right) + \frac{1}{3} - \frac{1}{4} = \frac{c}{60} + \frac{1}{12} = \frac{2}{15}, \\ DY &= \frac{2}{15} - \left(\frac{7}{24}\right)^2 = \frac{139}{2880} \approx 0.0483. \end{aligned}$$

Consequently,

$$Cor(X,Y) = \frac{\frac{11}{120} - \frac{5}{12}\frac{7}{24}}{\sqrt{\frac{43}{720}\frac{139}{2880}}} \approx -0.562, \qquad Cov(X,Y) = -\frac{43}{1440} \approx -0.030.$$

### 2. Conditional expectation

For $0 \le x \le 1/2$,

$$f_{X|Y}(x \mid 1/2) = \frac{f_{X,Y}(x, 1/2)}{f_Y(1/2)} = \frac{cx+1}{7/8}.$$

Thus

$$\mathbb{E}[X \mid Y = 1/2] = \int_0^{1/2} x\frac{cx+1}{7/8}\,dx = \frac{8}{7}\left(\frac{c}{24} + \frac{1}{8}\right) = \frac{2}{7} \approx 0.2857.$$

# Problem 2

1. From $\mathbb{P}(N(\mu, 10^2) < 250) = 15\%$, the standard normal table gives a quantile approximately $-1.04$ and the boxed result $\mu \approx 260.4$.

   *Transcription note: the handwritten standardisation line uses a denominator* $10^2$ *and then writes* $250 + 100 \cdot 1.04 = 260.4$. *The displayed arithmetic is inconsistent; the boxed mean is reproduced unchanged.*

2. $$\mathbb{P}(N(\mu, 10^2) > 280) = \mathbb{P}\left(N(0,1) > \frac{280-\mu}{10}\right) \approx 1 - \Phi(1.96) = \Phi(-1.96) \approx 0.025 = 2.5\%.$$

3. $$\begin{aligned}\mathbb{P}(230 \le N(250, \sigma^2) \le 270) &= 1 - 2\mathbb{P}(N(250, \sigma^2) < 230) \\ &= 1 - 2\Phi(-20/\sigma).\end{aligned}$$

   The notes use $\Phi(-20/\sigma) = 0.015$, hence $-20/\sigma \approx -2.17$ and $\sigma \approx 9.2166$.

# Problem 3

1. Let $X_i$ be student $i$'s score on problem 1. Then $X_i \sim 10 \cdot Ber(8/10)$, so

   $$\mathbb{E}X_i = 8, \qquad DX_i = 100\frac{8}{10}\frac{2}{10} = 16.$$

   With $S_n = X_1 + \cdots + X_n$, the CLT gives $(S_n - 8n)/(4\sqrt{n}) \xrightarrow{d} N(0,1)$. Thus

   $$\mathbb{P}(S_{100}/100 < 8) = \mathbb{P}\left(\frac{S_{100} - 800}{4\sqrt{100}} < 0\right) \approx \mathbb{P}(N(0,1) < 0) = \frac{1}{2}.$$

   The notes also suggest reasoning by symmetry and write $S_{100} \sim 10 \cdot Bin(100, 8/10)$.

2. Let $Y_i$ be the total points earned by student $i$, the sum of independent scores

   $$X_i^{(1)} \sim 10Ber(8/10), \quad X_i^{(2)} \sim 10Ber(7/10), \quad X_i^{(3)} \sim 10Ber(6/10), \quad X_i^{(4)} \sim 10Ber(4/10).$$

   Then $\mathbb{E}Y_i = 10(8/10 + 7/10 + 6/10 + 4/10) = 25$ and

   $$DY_i = 100\frac{8 \cdot 2 + 7 \cdot 3 + 6 \cdot 4 + 4 \cdot 6}{100} = 85.$$

   The grade is $Z_i = 2 + Y_i/10$, so $\mathbb{E}Z_i = 4.5$ and $DZ_i = DY_i/100 = 17/20$.

3. By the CLT,

   $$\frac{\frac{1}{n}\sum_{i=1}^n Z_i - 4.5}{\sqrt{17/20}/\sqrt{n}} \xrightarrow{d} N(0,1).$$

   Therefore

   $$\begin{aligned}\mathbb{P}\left(\frac{Z_1 + \cdots + Z_{100}}{100} - 4.5 > \frac{4.5}{100}\right) &= \mathbb{P}\left(\frac{\frac{1}{100}\sum Z_i - 4.5}{\sqrt{17/2000}} > \frac{4.5/100}{\sqrt{17/2000}}\right) \\ &\approx \mathbb{P}(N(0,1) > 0.49) \approx 31.21\%.\end{aligned}$$

# Problem 4

1. Let $X \sim Exp(1/100)$. The desired probability is

   $$\begin{aligned}\mathbb{P}(X < 200) + \mathbb{P}(X \in [300, 400]) &= 1 - \mathbb{P}(X \ge 200) + \mathbb{P}(X \ge 300) - \mathbb{P}(X \ge 400) \\ &= 1 - e^{-2} + e^{-3} - e^{-4} \approx 89.61\%,\end{aligned}$$

   using $\mathbb{P}(X > t) = e^{-\lambda t}$.

2. Ins 2 pays 200 leva with probability $\mathbb{P}(X > 300) = e^{-3}$. Its expected profit is

   $$e^{-3}(x - 200) + (1 - e^{-3})x = x - 200/e^3.$$

   Set this equal to 10, giving $x \approx 19.96$ leva.

3. Ins 1 pays $x = 10 + 200/e^3$ to Ins 2 and retains a claim cost $X$ if $X < 300$, or $X - 200$ if $X \geq 300$. Its expected retained claim is

$$\mathbb{E}X - 200\mathbb{P}(X \geq 300) = 100 - 200e^{-3} \approx 90.04 \text{ leva.}$$

If the policy price is $y$, require

$$y - \mathbb{E}[\text{total cost}] = 10, \qquad y - (x + 100 - 200/e^3) = 10,$$

so $y = 120$. The margin asks: "Can we say the answer is 120 regardless of the claim-size distribution?!" Without reinsurance, $y - \mathbb{E}[\text{claim}] = 10$, hence $y = 110$.